

- Functional Requirements Document (FRD)**
- Project Title:** Data Integration and Aggregation for Customer and Order Data
- Introduction:**

The purpose of this document is to outline the functional requirements for the data integration and aggregation process for customer and order data stored in CSV files.

- Functional Requirements:**

1. **Data Ingestion**

- The system shall read source CSV data from the specified volumes:
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`.
- The system shall load the data into Delta tables `customer` and `order`.

2. **Data Cleansing**

- The system shall remove "Null"/Null and duplicate records from both `customer` and `order` tables.

3. **Order Summary Table Creation**

- The system shall create an `ordersummary` table if it does not exist in
`catalog="gen\_ai\_poc\_databrickscoe"` and `schema="sdlc\_wizard"`.
- The schema of the `ordersummary` table shall be: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`,
`ItemName`, `PricePerUnit`, `Qty`, `Date`.

4. **Data Integration**

- The system shall join `customer` and `order` data using the `CustId` field.
- The system shall load the joined data into an SCD Type 2 table under
`catalog=gen\_ai\_poc\_databrickscoe`, `schema=sdlc\_wizard`, `table=ordersummary`.

5. **SCD Type 2 Table Maintenance**

- The system shall update the `ordersummary` table whenever there is a change in the `customer` table.
- The system shall make old records inactive and new records active, and update `StartDate` and `EndDate` accordingly.

6. **Customer Aggregate Spend Table Creation**

- The system shall create a `customeraggregatespend` table with columns `Name`, `TotalAmount`,
and `Date` in `catalog="gen\_ai\_poc\_databrickscoe"` and `schema="sdlc\_wizard"` if it does not exist.

7. **Data Aggregation**

- The system shall aggregate the `TotalAmount` column from the `ordersummary` table and group by the `Name` and `Date` columns.
- The system shall load the aggregated data into the `customeraggregatespend` table.
- Data Flow Diagram:**

```json

```
{
 "name": "generate_dfd_diagram",
 "arguments": {
 "text": "CSV Data ->> Data Ingestion: Read CSV data from volumes; Data Ingestion ->> Delta Tables: Load data into customer and order tables; Delta Tables ->> Data Cleansing: Remove Null/Null and duplicate records; Data Cleansing ->> Data Integration: Cleaned customer and order data; Data Integration ->> Order Summary Table: Join customer and order data using CustId; Order Summary Table ->> SCD Type 2 Table Maintenance: Update ordersummary table on customer data change; Order Summary Table ->> Data Aggregation: Aggregate TotalAmount by Name and Date; Data Aggregation ->> Customer Aggregate Spend Table: Load aggregated data into customeraggregatespend table;"
 }
}
```

```

The above JSON call will generate a DFD diagram illustrating the data flow from CSV data to the various tables.

- System Components:**

1. Data Ingestion Module
2. Data Cleansing Module
3. Data Integration Module
4. Order Summary Table Creation Module
5. SCD Type 2 Table Maintenance Module
6. Customer Aggregate Spend Table Creation Module
7. Data Aggregation Module

- Acceptance Criteria:**

1. The system successfully ingests data from the input CSV files.
2. The system performs data cleansing correctly.
3. The `ordersummary` table is created with the correct schema.
4. The system joins `customer` and `order` data correctly.
5. The `ordersummary` table is updated correctly when there are changes in the `customer` table.

6. The `customeraggregatespend` table is created with the correct schema.

7. The system aggregates data correctly and loads it into the `customeraggregatespend` table.

By following this FRD, the development team should be able to design and implement a data integration and aggregation process that meets the business requirements.